

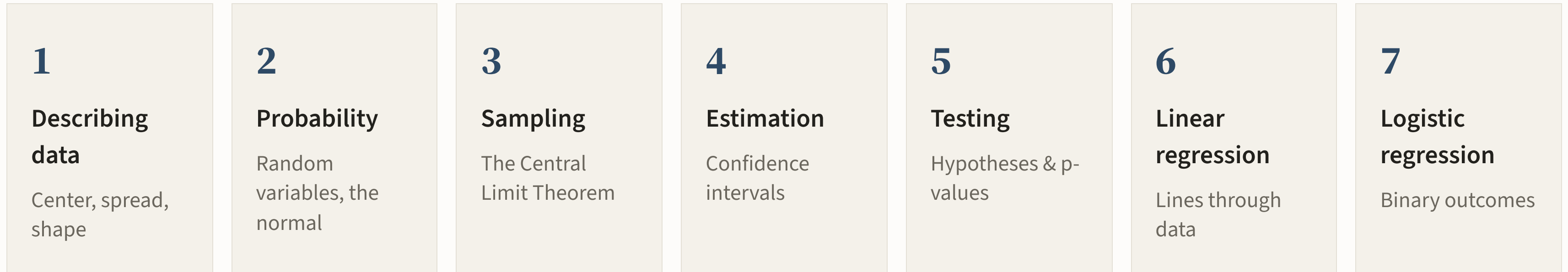
UNDERGRADUATE LECTURE · DATA ANALYSIS

Foundations of Data Analysis

From describing a dataset to testing hypotheses and fitting regression models — one step at a time.

≈ 3 hours · 7 parts · One running example

From Data to Models



Every part reuses the vocabulary of the one before it. If a later slide feels confusing, the answer is almost always two parts back.

The Campus Café

A café near the library recorded **one row per day** for a full semester — 112 days of operations.

Every method in this course will be a different question asked of this same table.

```
import pandas as pd
df = pd.read_csv("cafe.csv")
df.shape  # (112, 6)
```

Column	Meaning
date	Calendar day
cups_sold	Total drinks sold
iced_sold	Iced drinks sold
temp	Daily high, °C
promo	Promotion running? yes / no
sold_out	Ran out of stock? yes / no

PART 1 OF 7

Describing Data

Center, spread, and shape — the vocabulary for everything that follows.

Types of Variables

Numerical

Values you can meaningfully average and subtract.

cups_sold

iced_sold

temp

Summaries: mean, median, variance, histograms.

Categorical

Labels that place each day into a group.

promo

sold_out

Summaries: counts, proportions, bar charts.

This one distinction will route us later: numerical outcome → **linear regression**; binary categorical outcome → **logistic regression**.

Measures of Center

Mean

$$\bar{X} = \frac{1}{n} \sum x_i$$

Add everything up, divide by how many. Uses **every value** — so extreme days pull it around.

Median

the middle value

Sort the values; take the one in the middle. Uses **only the order** — so extreme days barely matter.

```
df["cups_sold"].mean()    # 236.0  
df["cups_sold"].median()  # 234.5
```

Measures of Spread

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

1. Measure each day's **deviation** from the mean.
2. **Square** it — so ups and downs don't cancel out.
3. Average the squares. The result is the **variance**.
4. Take the square root: the **standard deviation s**, back in cups.

RULE OF THUMB

For bell-shaped data, about **68%** of days fall within **1 SD** of the mean, and about **95%** within **2 SD**.

We'll see where these numbers come from in Part 2.

WORKED EXAMPLE

Ten Days of Sales

212	231	198	245	260	224	251	205	238	296
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Mean

$2,360 / 10 = \mathbf{236}$

cups per day

Median

$(231+238)/2 = \mathbf{234.5}$

middle of the sorted list

Standard deviation

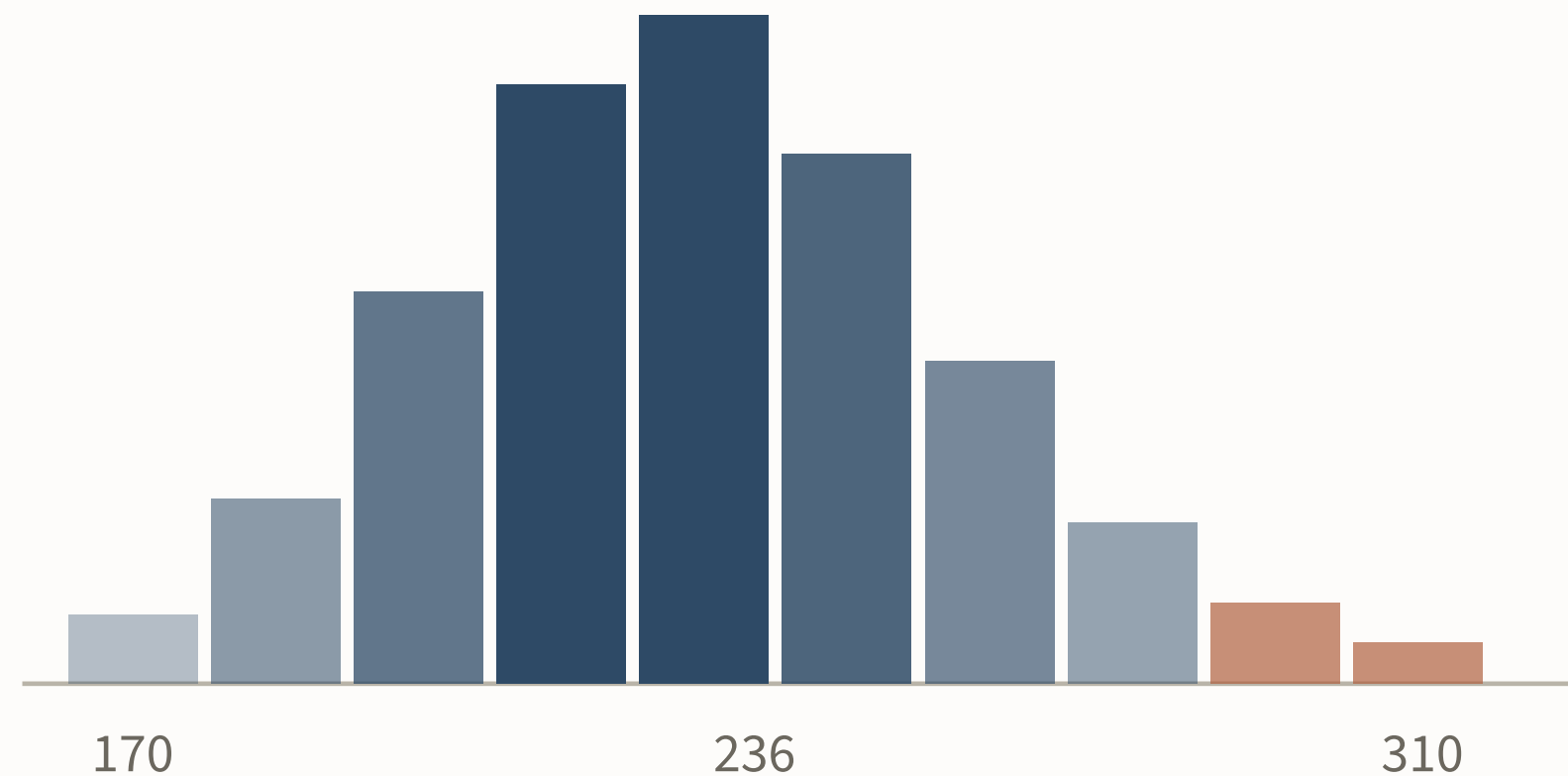
$\sqrt{(7,636 / 9)} \approx \mathbf{29.1}$

$\Sigma(x_i - 236)^2 = 7,636 \cdot \text{divide by } n - 1$

Read it as a sentence: “A *typical day sells about 236 cups, give or take about 29.*” Note the **296-cup day** pulls the mean above the median.

Histograms & Distribution Shape

cups_sold, 112 days



```
df["cups_sold"].hist(bins=10)
```

- **Center:** the tall bars sit near 236.
- **Spread:** most days land between ~180 and ~290.
- **Shape:** a slight **right tail** — a few unusually busy days.

Shape is the thing numbers alone can't show. Always plot before you summarize.

COMMON MISCONCEPTION

“The Mean Is Always Typical”

THE TRAP

Suppose 9 days sell ~200 cups and one festival day sells 2,000. The mean is **380 cups** — a number that describes *no actual day*.

THE FIX

Check the shape first. If the histogram is skewed, report the **median** alongside — or instead of — the mean.

In skewed data the mean chases the tail: **mean > median** under right skew, **mean < median** under left skew. The gap between them is itself a diagnostic.

Describing Data

- 1** Every numerical variable gets three descriptions: **center** (mean, median), **spread** (SD), and **shape** (histogram).
- 2** The mean uses every value and follows outliers; the median resists them. Skew pushes them apart.
- 3** The standard deviation is the “give or take” number — same units as the data, and the currency of everything ahead.

PART 2 OF 7

Probability Foundations

Random variables, expectation, and the normal distribution — the machinery of uncertainty.

Random Variables

A **random variable** is a number whose value is decided by chance. Before tomorrow happens, tomorrow's sales are one:

$X = \text{cups sold tomorrow}$

Its **distribution** lists the possible values and how likely each one is. Each observed day is one **draw** from that distribution.

THE KEY SHIFT

Part 1: the histogram of days we *saw*.

Part 2: the distribution of days that *could happen*.

The observed histogram is our best glimpse of the true, unseen distribution.

Expectation & Variance

Expectation

$$E[X] = \mu$$

The long-run average over infinitely many days — a probability-weighted mean of all possible values.

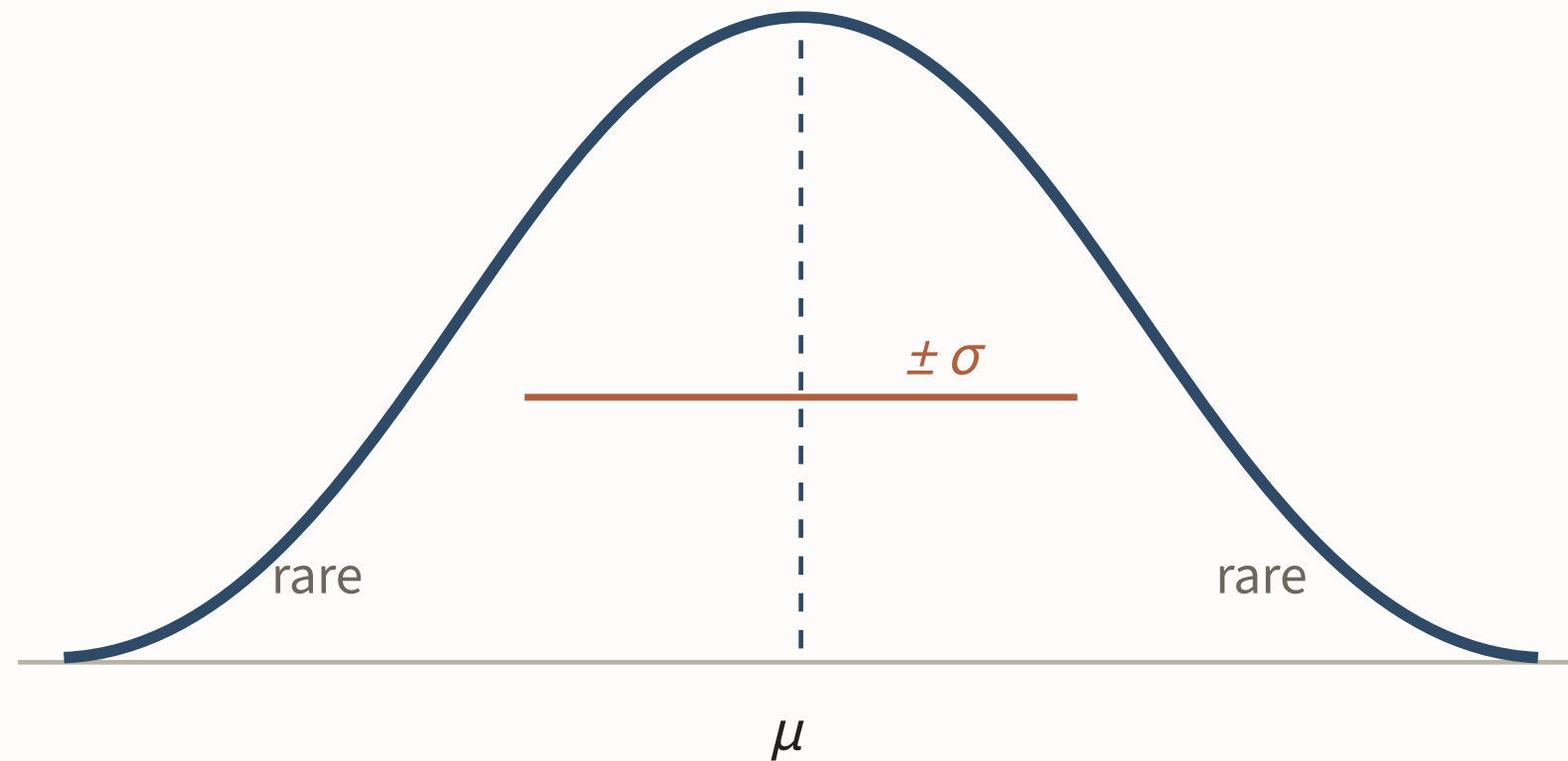
Variance

$$\text{Var}(X) = E[(X - \mu)^2] = \sigma^2$$

The expected squared deviation — how much draws typically wobble around μ .

The naming convention of the whole course: Greek letters (μ , σ) are the *true but unknown* quantities. Latin letters ($\bar{x}$, s) are what we *compute from data* to estimate them.

The Normal Distribution



$$X \sim N(\mu, \sigma^2)$$

Symmetric, bell-shaped, and fully described by just **two numbers**: the center μ and the spread σ .

within 1σ of μ	68%
within 2σ of μ	95%
within 3σ of μ	99.7%

Why it appears everywhere: Part 3 will give the answer.

z-Scores · Standardizing Values

$$Z = \frac{X - \mu}{\sigma}$$

Subtract the center, divide by the spread. The result: **how many standard deviations** a value sits from the mean.

Standardizing maps *every* normal distribution onto one shared ruler: **N(0, 1)**.

READING Z

$z \approx 0$ an ordinary value

$|z| \approx 1$ common wobble

$|z| > 2$ unusual — top/bottom ~2.5% each

Keep “ $|z| > 2$ is unusual” in mind — hypothesis testing is built on exactly this idea.

WORKED EXAMPLE

How Unusual Was the 296-Cup Day?

Suppose daily sales are roughly normal with $\mu = 236$ and $\sigma = 29$. One day the café sells **296 cups**.

Step 1 · Deviation

$$296 - 236 = \mathbf{60}$$

cups above the mean

Step 2 · Standardize

$$60 / 29 \approx \mathbf{2.07}$$

standard deviations above

Step 3 · Interpret

$$\text{top} \approx \mathbf{2\%}$$

$$P(Z > 2.07) \approx 0.019$$

```
from scipy import stats
1 - stats.norm.cdf(296, loc=236, scale=29)  # 0.019
```

Probability Foundations

- 1** Data are **draws from a distribution**. Greek letters (μ , σ) name the true distribution; Latin letters ($\bar{x}$, s) name our estimates.
- 2** The normal distribution is fully pinned down by μ and σ , with the 68 / 95 / 99.7 rule for areas.
- 3** A z-score measures distance from the mean in SD units. **$|z| > 2 \approx \text{unusual}$** — the seed of hypothesis testing.

PART 3 OF 7

Sampling & the Central Limit Theorem

Why averages behave so much more predictably than the data they came from.

Population vs. Sample

Population

All café days there could ever be — this semester, next semester, every semester. Mean μ , SD σ .

We never observe it directly.

Sample

The $n = 36$ weekdays we happened to record. Mean $\bar{x} = 236$, SD $s = 29.1$.

This is all we ever get.

The central question of statistics: what does $\bar{x} = 236$, computed from 36 days, let us say about the unknown μ ?

The Sampling Distribution of the Mean

A thought experiment:

1. Rerun the semester. Record 36 new days. Compute a new $\bar{x}$.
2. Do it again. And again — thousands of times.
3. Collect all those $\bar{x}$'s into a histogram.

That histogram is the **sampling distribution of $\bar{x}$** — the distribution of the estimate itself.

WHY IT MATTERS

Our single $\bar{x} = 236$ is **one draw** from this distribution. If we know how much $\bar{x}$ wobbles across reruns, we know how far it plausibly sits from μ .

We never actually rerun the semester — the next slide says we don't have to.

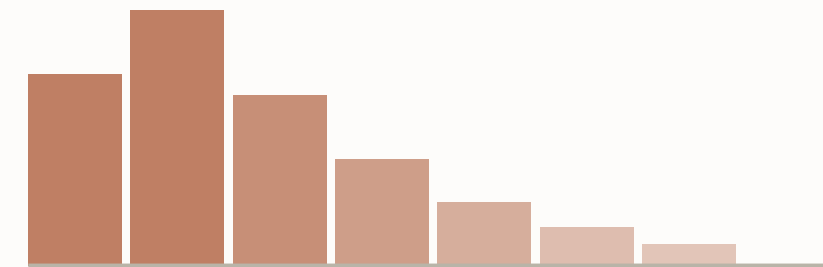
The Central Limit Theorem

$$\bar{X} \approx N(\mu, \sigma^2/n)$$

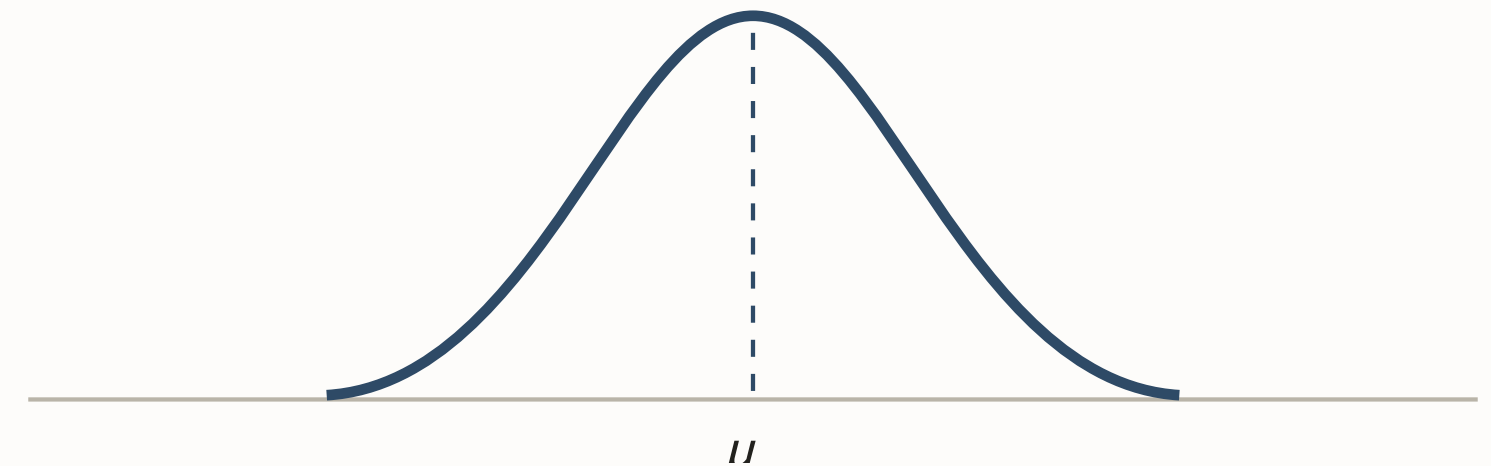
For large enough n , the sample mean is approximately **normal** — centered on μ , with variance shrunk by a factor of n — **no matter what shape** the raw data have.

```
# simulate: means of skewed data look normal
means = [rng.exponential(2, 36).mean()
         for _ in range(10_000)]
plt.hist(means, bins=50) # bell-shaped!
```

raw data: skewed



means of samples ($n = 36$): bell-shaped, narrow



Standard Error

$$SE = \frac{s}{\sqrt{n}}$$

The **standard deviation of $\bar{x}$ itself** — how much the sample mean would wobble across repeated samples.

For the café: $SE = 29.1 / \sqrt{36} = \mathbf{4.85 \text{ cups}}$. The mean of 36 days is far steadier than any single day.

THE $\sqrt{N}$ LAW

$$n = 9 \qquad SE = 29.1 / 3 \approx \mathbf{9.7}$$

$$n = 36 \qquad SE = 29.1 / 6 \approx \mathbf{4.85}$$

$$n = 144 \qquad SE = 29.1 / 12 \approx \mathbf{2.4}$$

Halving uncertainty costs 4× the data. Precision is expensive.

COMMON MISCONCEPTION

“SD and SE Are the Same Thing”

Standard deviation s

Spread of the **data**: how much individual days differ from each other.

Collecting more days **does not shrink it** — days stay as variable as they are.

Standard error $s/\sqrt{n}$

Spread of the **estimate**: how much $\bar{x}$ would vary across repeated samples.

Collecting more days **shrinks it** — averages stabilize as n grows.

Quick check: “the SD of café sales is 29 cups” vs. “our estimate of the mean is precise to about ± 5 cups.” Different questions, different numbers.

Sampling & the CLT

- 1** An estimate like $\bar{x}$ has its **own distribution** — the sampling distribution — describing how it varies across possible samples.
- 2** The CLT: for large n , that distribution is **normal** — whatever the raw data look like. This is why the normal curve is everywhere.
- 3** Its spread is the **standard error $s/\sqrt{n}$** — the single most important formula of the course.

PART 4 OF 7

Estimation & Confidence Intervals

From a single number to an honest range of plausible values.

Point Estimates & Uncertainty

Our best single guess for μ is the **point estimate** $\bar{x} = 236$.

But a different 36 days would have given 233, or 239, or 241... The estimate is **almost surely not exactly right**.

The honest report is a **range**: the estimate, plus or minus its typical error.

THE RECIPE (FROM THE CLT)

$\text{estimate} \pm \text{margin of error}$

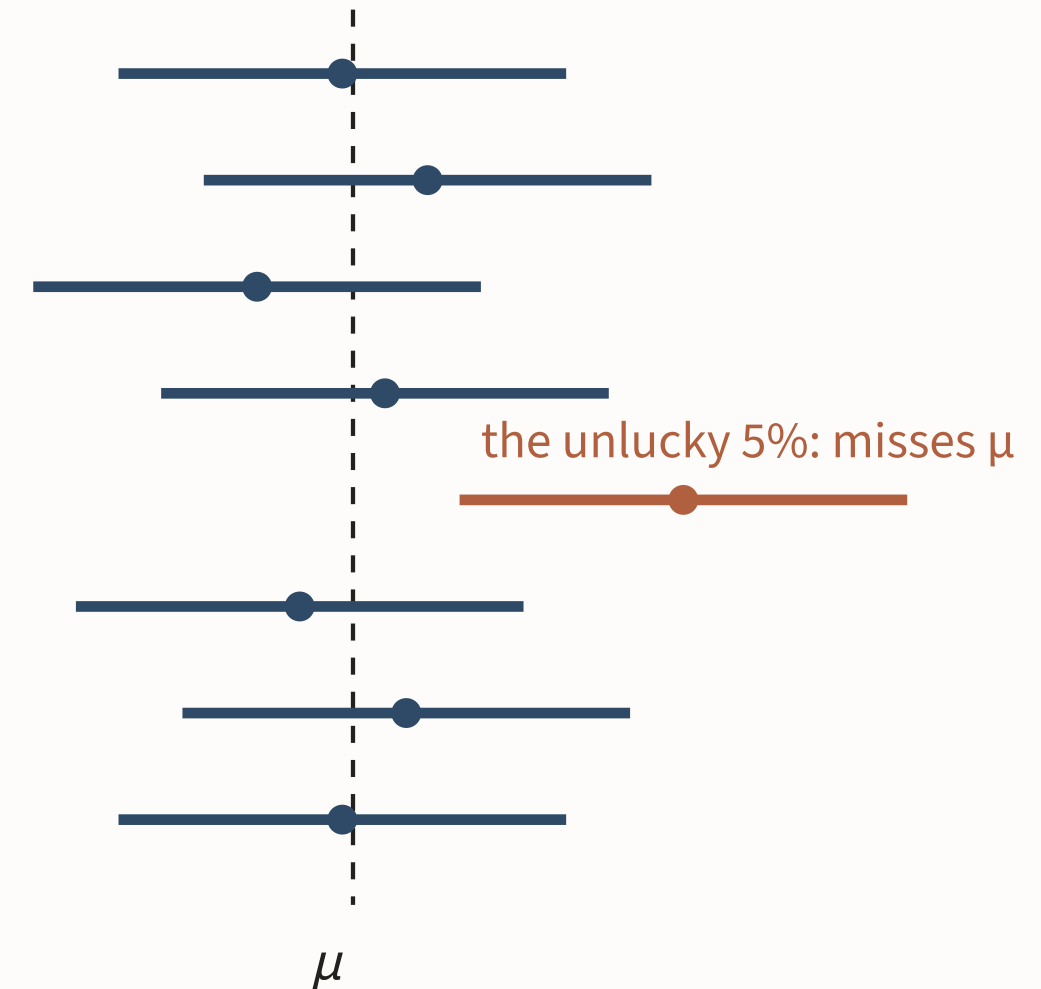
where the margin is a few **standard errors** —
because the CLT says $\bar{x}$ lands within ~ 2 SE of μ about 95% of the time.

The 95% Confidence Interval

$$\bar{x} \pm t^* \cdot \frac{s}{\sqrt{n}}$$

$t^* \approx 2$ for a 95% interval. The interval is the set of values of μ that are **consistent with the data**.

“95%” describes the **procedure**: across many studies, intervals built this way capture the truth 95% of the time.

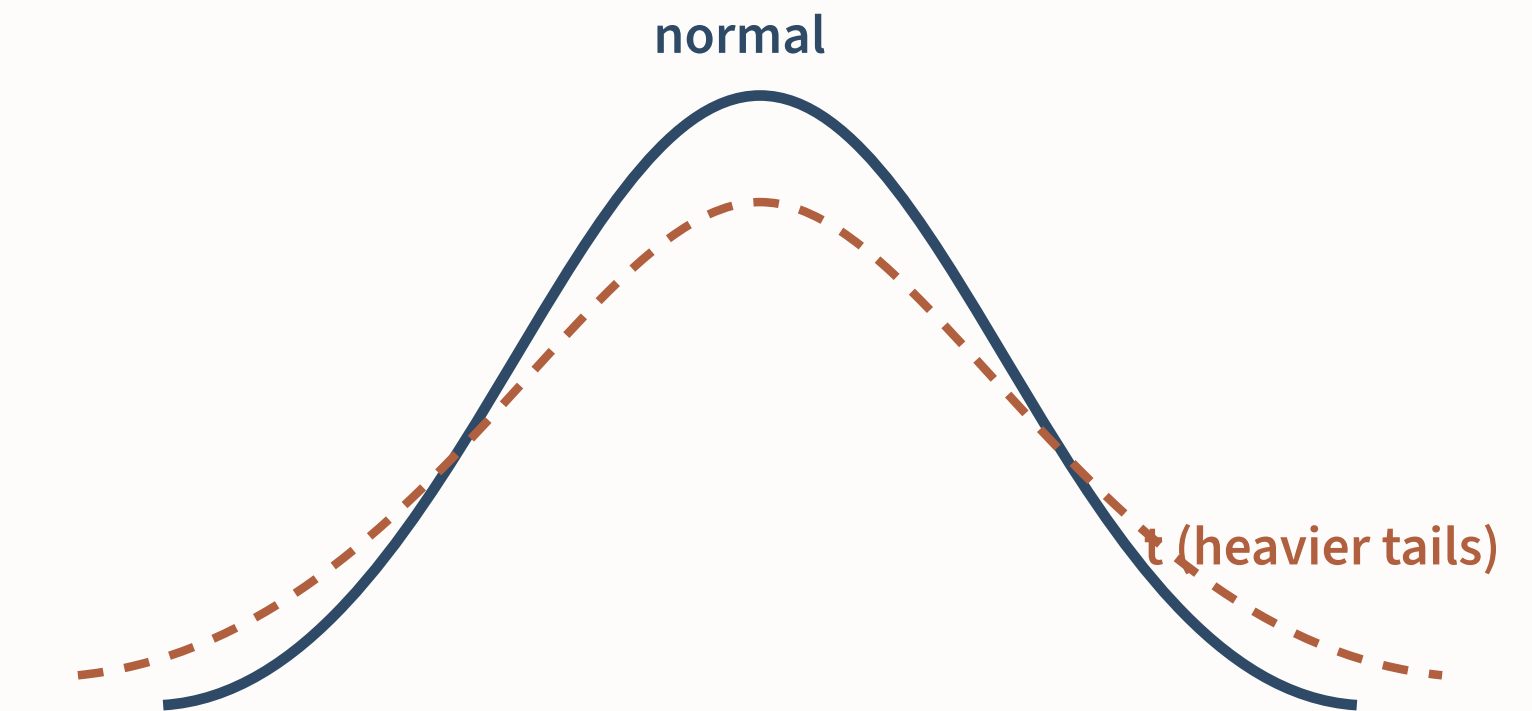


The t-Distribution

The CLT uses the true σ — but we only have the estimate s . That substitution adds uncertainty.

The **t-distribution** is the normal curve's cautious cousin: same shape, **heavier tails**, governed by degrees of freedom **$df = n - 1$** .

$df = 5$	$t^* = \mathbf{2.57}$
$df = 35$	$t^* = \mathbf{2.03}$
$df \rightarrow \infty$	$t^* = \mathbf{1.96}$ (the normal)



```
stats.t.ppf(0.975, df=35)  # 2.03
```


WORKED EXAMPLE

A 95% CI for Mean Daily Sales

Ingredients

$$\bar{x} = 236$$

$$s = 29.1, n = 36$$

Margin of error

$$\begin{aligned} &2.03 \times 29.1/\sqrt{36} \\ &= 2.03 \times 4.85 \approx \mathbf{9.8} \end{aligned}$$

Interval

$$\begin{aligned} &236 \pm 9.8 \\ &= \mathbf{(226.2, 245.8)} \end{aligned}$$

The written conclusion: “*We are 95% confident that the café's true mean daily sales lie between 226 and 246 cups.*”

```
stats.t.interval(0.95, df=35, loc=236, scale=4.85)
```

COMMON MISCONCEPTION

“There's a 95% Chance μ Is in My Interval”

WHY IT'S OFF

μ is a fixed number, not a coin flip. Once (226.2, 245.8) is computed, μ is either in it or not — there is no 95% left.

THE CORRECT READING

The **method** captures μ in 95% of studies. Our confidence is in the recipe — we hold one ticket from a lottery where 95% of tickets win.

In practice, the pragmatic reading — “values in the interval are plausible for μ ; values far outside are not” — serves you well.

Estimation & Confidence Intervals

- 1 Never report an estimate alone. The universal pattern is **estimate $\pm t^* \times SE$** .
- 2 Use **t** instead of the normal because s stands in for σ ; the difference fades as n grows.
- 3 “95% confident” is a statement about the **procedure's success rate**, not about any single interval.

PART 5 OF 7

Hypothesis Testing

A formal procedure for asking: is this claim consistent with the data?

The Logic of a Hypothesis Test

1

Assume the skeptic is right

“Nothing changed” — the null hypothesis.

2

Look at the data

Measure how far it sits from what the null predicts.

3

Ask: coincidence?

How often would chance alone produce data this extreme?

4

Decide

Rarely → reject the null. Often → the data prove nothing new.

This is proof by contradiction, softened with probability: we can never be certain, but we can measure how surprising the evidence is.

Null & Alternative Hypotheses

The café's question: last year the mean was 225 cups/day. After a menu redesign, has it changed?

Null $H_0 : \mu = 225$

The skeptic's position: **nothing changed**. Any difference in our sample is just sampling noise.

Always contains “=” — it must specify an exact world we can compute probabilities in.

Alternative $H_1 : \mu \neq 225$

The claim that needs evidence: the redesign **moved the mean**, in either direction.

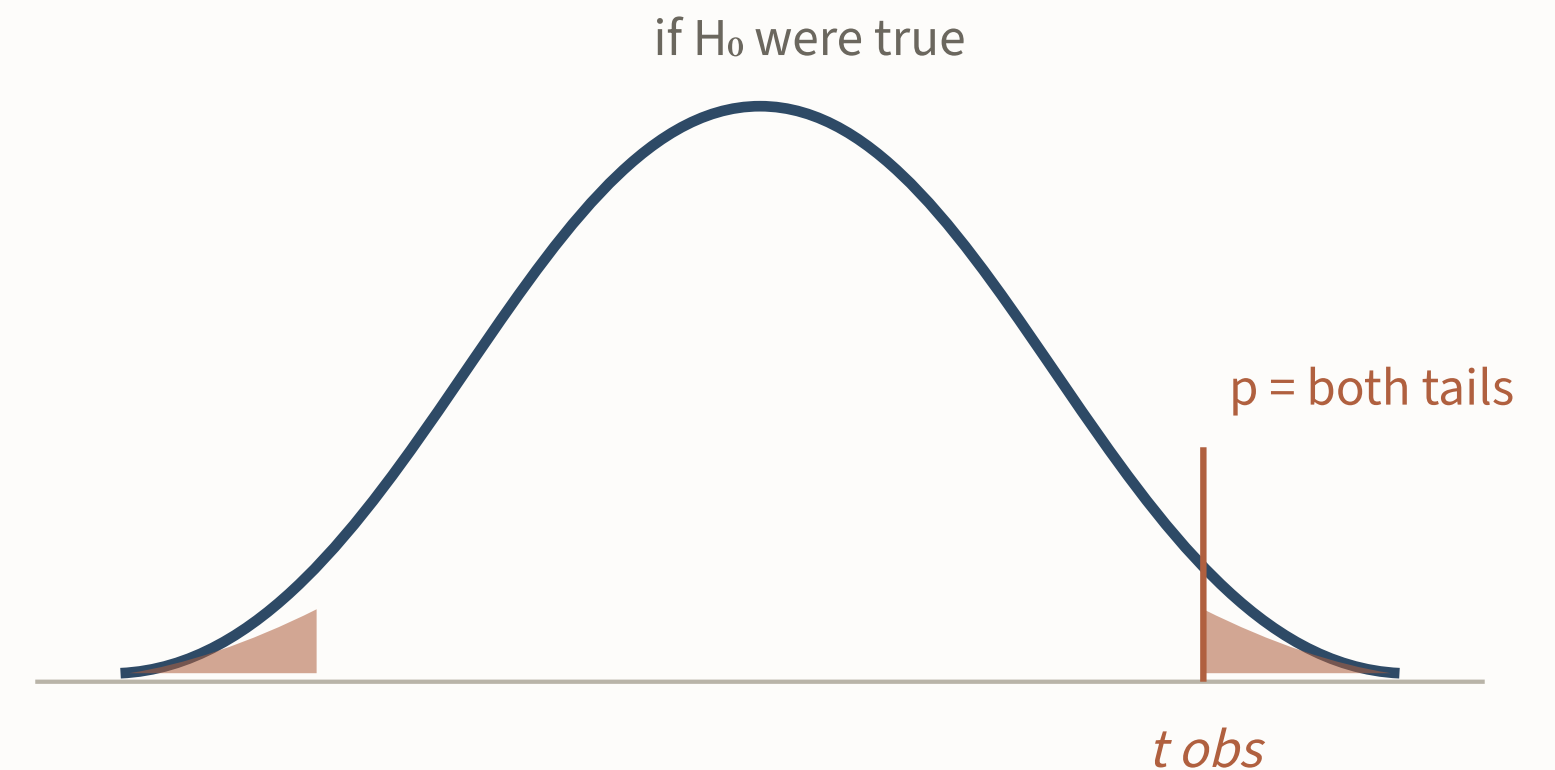
The burden of proof sits here. We only accept it if the data force us to.

Test Statistics & p-Values

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

Recognize it? It's a **z-score for $\bar{x}$** , computed in the world where H_0 is true. It counts standard errors between the data and the null.

The **p-value** translates t into a probability: *if H_0 were true, how often would we see a result at least this extreme?*



WORKED EXAMPLE

Did Sales Change After the Redesign?

Hypotheses

$$H_0: \mu = 225$$

$$H_1: \mu \neq 225$$

Test statistic

$$t = (236 - 225) / 4.85 \\ = \mathbf{2.27}$$

p-value (df = 35)

$$p \approx \mathbf{0.029}$$

$< 0.05 \rightarrow$ reject H_0

Conclusion: *“If nothing had changed, a sample mean this far from 225 would occur in only ~3% of semesters. The evidence suggests sales genuinely increased.”* Note the CI agrees: 225 falls outside (226.2, 245.8).

```
stats.ttest_1samp(df["cups_sold"], popmean=225)
```


Comparing Two Groups

New question: do **promo days** sell more than regular days?

Promo days (n = 41)	$\bar{x} = \mathbf{247}$
Regular days (n = 71)	$\bar{x} = \mathbf{231}$
Difference	+16 cups , SE ≈ 5.9

Same logic, new estimate: **$t = 16 / 5.9 \approx 2.7$** , $p \approx 0.008 \rightarrow$ promo days really do differ.

THE UNIVERSAL SKELETON

$$t = \frac{\text{estimate} - \text{null value}}{\text{standard error}}$$

One mean, a difference of means, a regression slope — every t-test in this course is this one fraction with different ingredients.

Type I & Type II Errors

	H ₀ actually true	H ₀ actually false
We reject H ₀	Type I error — false alarm. Probability = α (we choose it, usually 0.05).	Correct — a real effect, detected.
We fail to reject	Correct — no effect, none claimed.	Type II error — a real effect, missed. Shrinks with larger n.

$\alpha = 0.05$ is a convention, not physics: it accepts one false alarm in twenty as the price of being able to detect real effects.

COMMON MISCONCEPTION

“ $p = 0.029$ Means H_0 Has a 2.9% Chance of Being True”

WHAT P IS NOT

- ✗ the probability H_0 is true
- ✗ the probability the result is a fluke
- ✗ a measure of how *big* or important the effect is

WHAT P IS

Assuming H_0 is true, the probability of data at least as extreme as ours.

It measures **surprise under the null** — nothing more.
A tiny effect with huge n can have a tiny p .

Always report the **effect size with its CI** next to the p -value: “+11 cups, 95% CI (1.2, 20.8), $p = 0.029$.”

Hypothesis Testing

1 A test assumes the null, then measures surprise: **$t = (\text{estimate} - \text{null}) / \text{SE}$** , converted to a p-value.

2 p is the probability of the **data given the null** — never the probability of the null itself.

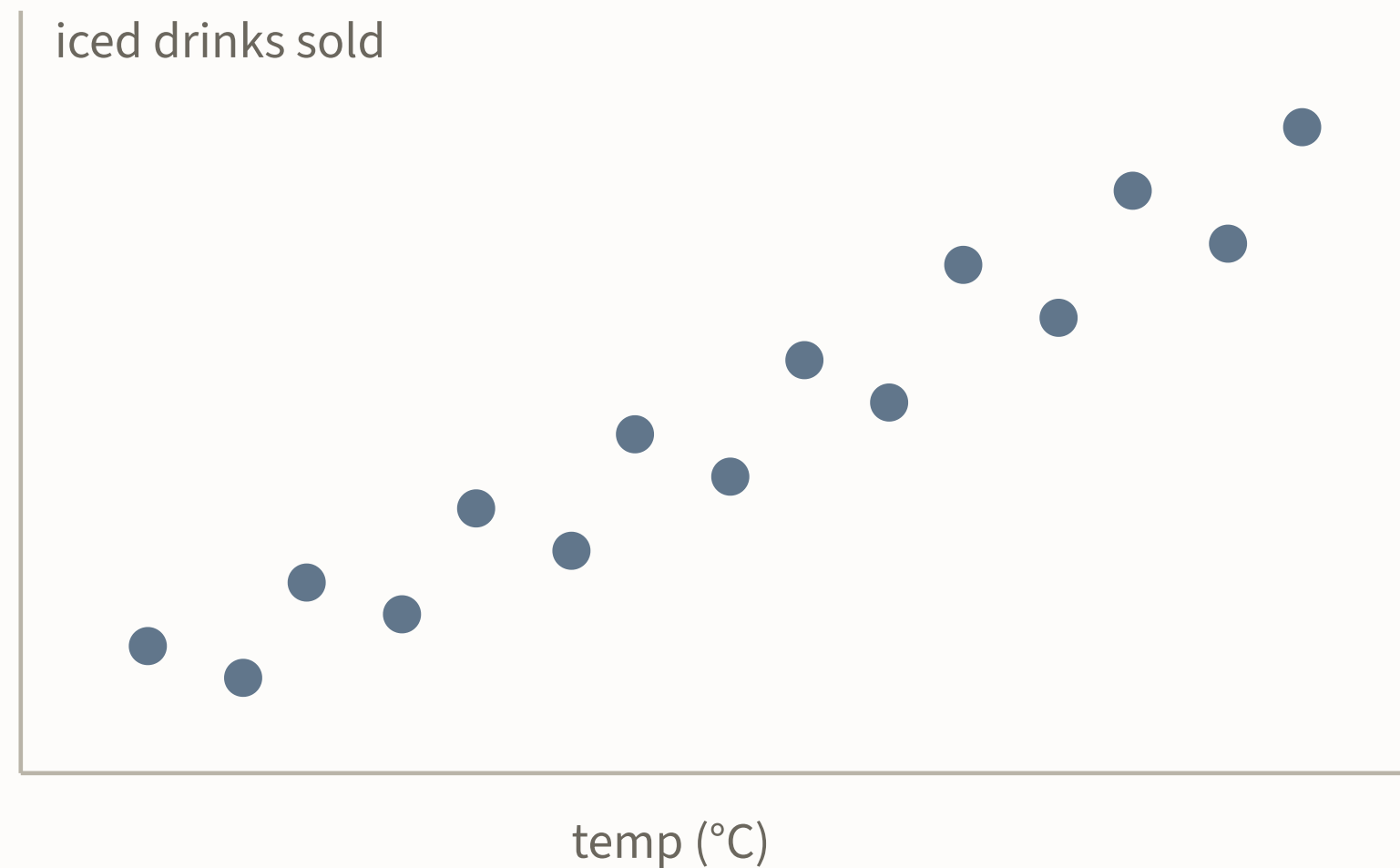
3 Significance $\neq$ importance. Report the **effect size and its CI** alongside every p-value.

PART 6 OF 7

Linear Regression

Modeling how a numerical outcome responds to a predictor.

Scatterplots & Correlation



Hotter days, more iced drinks — the cloud of points **tilts upward**.

The **correlation coefficient r** scores the tilt's tightness, from -1 to $+1$. Here: **$r = 0.76$** — strong and positive.

$$|r| \approx 0$$

no linear pattern

$$|r| \approx 0.5$$

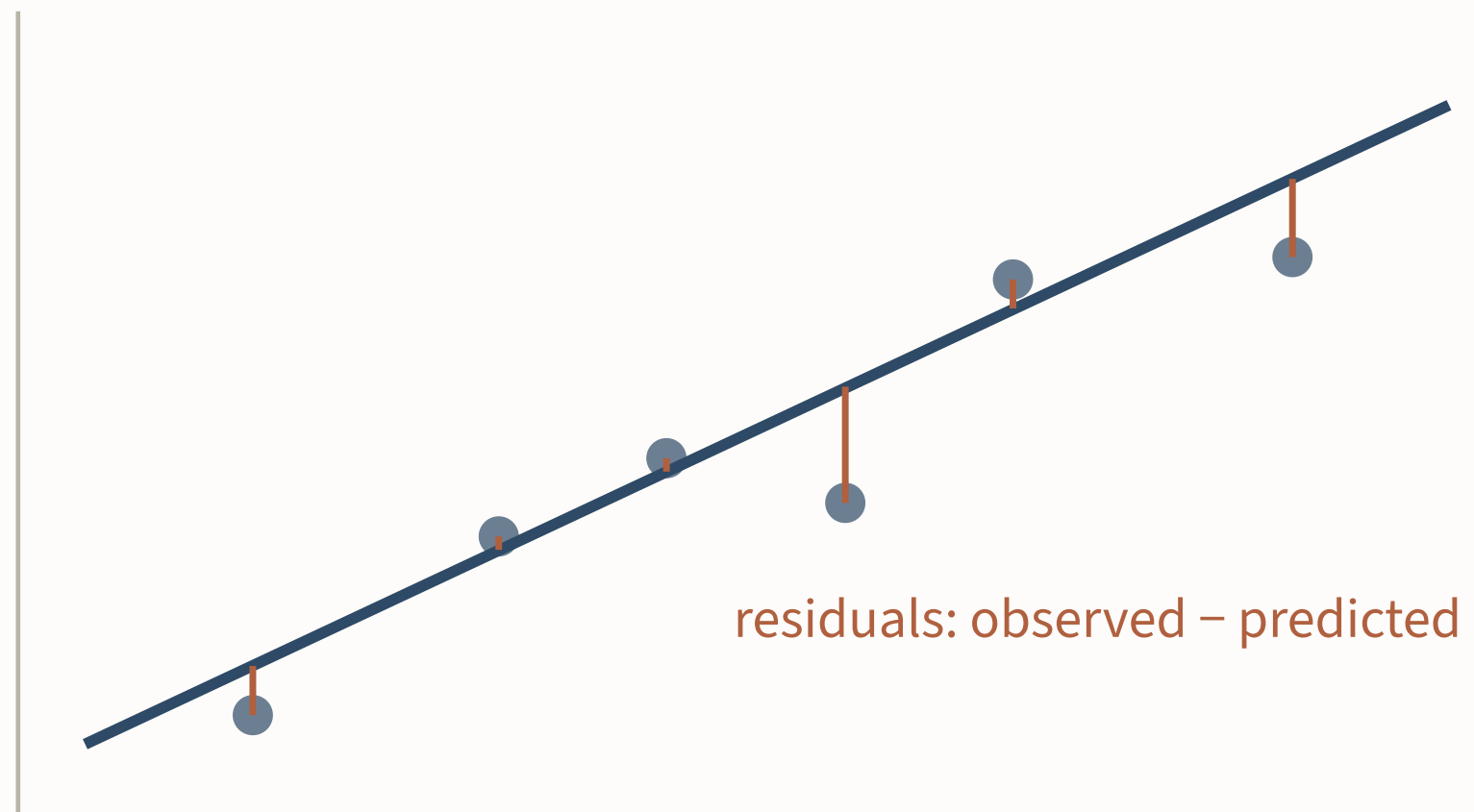
visible but loose

$$|r| \approx 1$$

points hug a line

r sees only **linear** patterns. Curves, clusters, and outliers hide from it — plot first, always.

The Least-Squares Line



$$\hat{y} = b_0 + b_1x$$

Of all possible lines, least squares picks the one minimizing the **sum of squared residuals**:

$$\min \sum (y_i - \hat{y}_i)^2$$

The same move as variance: square the misses so over- and under-predictions can't cancel.

Solving for the Coefficients

“Minimize $\sum(y_i - b_0 - b_1x_i)^2$ ” is a calculus problem: take the derivative with respect to each coefficient, set both to zero. Two equations, two unknowns — and a **closed-form solution**:

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = r \cdot \frac{s_y}{s_x}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

CAFÉ NUMBERS

$r = 0.76$, $s_y = 22.9$ iced drinks, $s_x = 6.0$ °C

$b_1 = 0.76 \times 22.9 / 6.0 \approx \mathbf{2.9}$

$b_0 = 43 - 2.9 \times 21 \approx \mathbf{-18}$

Read the slope formula: it is the correlation r , re-scaled from “SDs of y per SD of x ” into real units. And b_0 simply forces the line through the point of means $(\bar{x}, \bar{y})$.

Interpreting Slope & Intercept

$$\widehat{\text{iced}} = -18 + 2.9 \times \text{temp}$$

Slope $b_1 = 2.9$

Each additional degree is **associated with** about 2.9 more iced drinks per day. The unit: *outcome per unit of predictor*.

Intercept $b_0 = -18$

The prediction at $\text{temp} = 0^\circ\text{C}$ — a freezing day. Negative drinks are impossible: the intercept **anchors the line**, it need not be meaningful on its own.

Prediction in action: a 30°C day $\rightarrow -18 + 2.9 \times 30 \approx \mathbf{69 \text{ iced drinks}}$. Beware extrapolating far beyond the observed 8–33 $^\circ\text{C}$ range.

Inference on the Slope

The slope 2.9 was computed from **one sample** of days — a new semester would give a slightly different slope. Sound familiar?

So we reuse the entire Part 4–5 machinery on b_1 :

$$t = \frac{b_1 - 0}{SE(b_1)} = \frac{2.9}{0.42} \approx 6.9$$

$p < 0.001$, and the 95% CI is **(2.1, 3.7)** — solidly away from zero.

WHY NULL = 0?

A slope of exactly 0 means the line is flat:

temperature carries no information about iced sales. Rejecting it establishes there is a relationship at all.

```
import statsmodels.formula.api as smf
smf.ols("iced_sold ~ temp",
        data=df).fit().summary()
```

Model Fit • R²

$$R^2 = \frac{\text{variance explained}}{\text{total variance}} = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

- $SST = \Sigma(y_i - \bar{y})^2$

total sum of squares — variation of y around its mean
- $SSR = \Sigma(\hat{y}_i - \bar{y})^2$

regression sum of squares — variation the line accounts for
- $SSE = \Sigma(y_i - \hat{y}_i)^2$

error sum of squares — the squared residuals

Least squares guarantees **SST = SSR + SSE**. For the café: **R² = 0.58** — temperature accounts for 58% of the variation in iced sales.

With one predictor, R² is just r² : 0.76² ≈ 0.58. Divide each sum by n – 1 for variances — the ratio is unchanged.



A low R² does not mean the effect is fake — the slope can be real and precisely estimated while much of life stays unexplained. Fit and significance are different questions.

COMMON MISCONCEPTION

“A Significant Slope Proves X Causes Y”

THE TRAP

Suppose `iced_sold` also predicts sunburn cases on campus. Iced coffee doesn't burn anyone — **sunny weather drives both**.

A lurking third variable (a *confounder*) can manufacture any correlation.

WHAT EARNS A CAUSAL CLAIM

Randomized experiments — or careful adjustment for confounders plus a plausible mechanism. Regression alone describes **association**, and that is already useful: for prediction, no causation is required.

Say “associated with,” not “causes” — the phrasing on the slope slide was chosen deliberately.

Linear Regression

- 1** Least squares finds the line minimizing squared residuals; the **slope** is the expected change in y per unit of x.
- 2** Inference on the slope reuses the universal skeleton: $\mathbf{t} = \mathbf{b}_1 / \mathbf{SE}(\mathbf{b}_1)$, with null slope 0 meaning “no relationship.”
- 3** R^2 measures fit, the p-value measures evidence, and **neither one proves causation.**

PART 7 OF 7

Logistic Regression

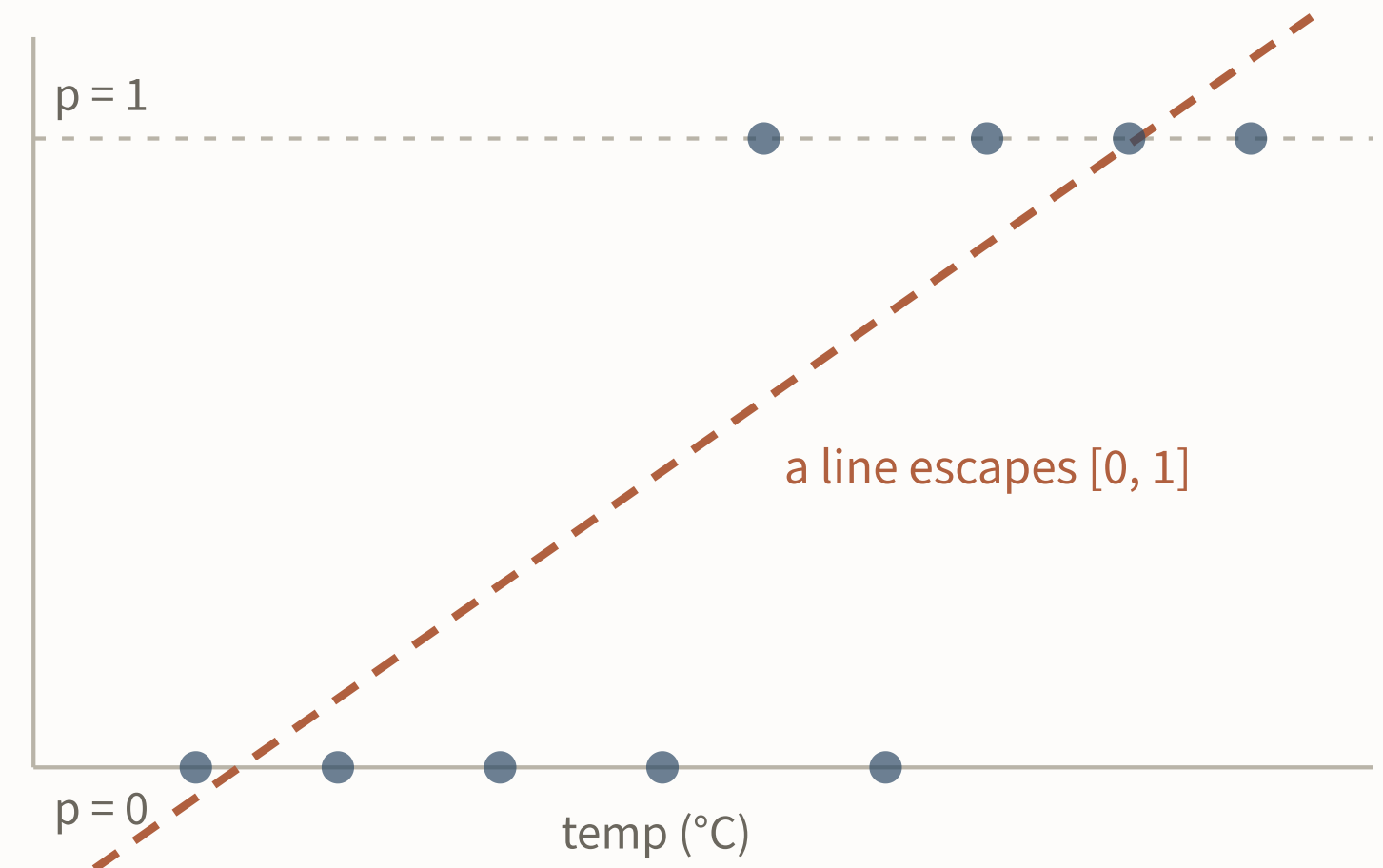
Regression when the outcome is yes or no.

When the Outcome Is Binary

New question: on which days does the café **sell out**? The outcome isn't a count — it's $\text{sold_out} \in \{\text{no}, \text{yes}\}$.

What we model is the **probability**: $p = P(\text{sold out} \mid \text{temp})$.

A straight line fails: extend it far enough and it predicts probabilities **below 0 or above 1** — nonsense we must design out.



The Bernoulli Distribution

A yes/no outcome has the simplest distribution there is.

Each day i is one **Bernoulli trial**:

$$Y_i \sim \text{Bernoulli}(p_i)$$

$P(Y_i = 1) = p_i$, $P(Y_i = 0) = 1 - p_i$ — a coin flip whose bias p_i **depends on that day's temperature**. Logistic regression is the model for how.

TWO FACTS WE'LL USE

$E[Y_i] = p_i$ the mean of a 0/1 variable *is* its probability

$\text{Var}(Y_i) = p_i(1 - p_i)$ largest at $p = 0.5$, zero at the certain extremes

The parallel to Part 6: linear regression models $E[Y]$ as $b_0 + b_1x$. Here $E[Y] = p$ — so we are again modeling a mean, just one trapped in $[0, 1]$.

Odds & Log-Odds

Probability

p

lives in $[0, 1]$ — too cramped for a line

Odds

$p / (1 - p)$

lives in $[0, \infty)$ — “3 to 1” in gambling speak

Log-odds

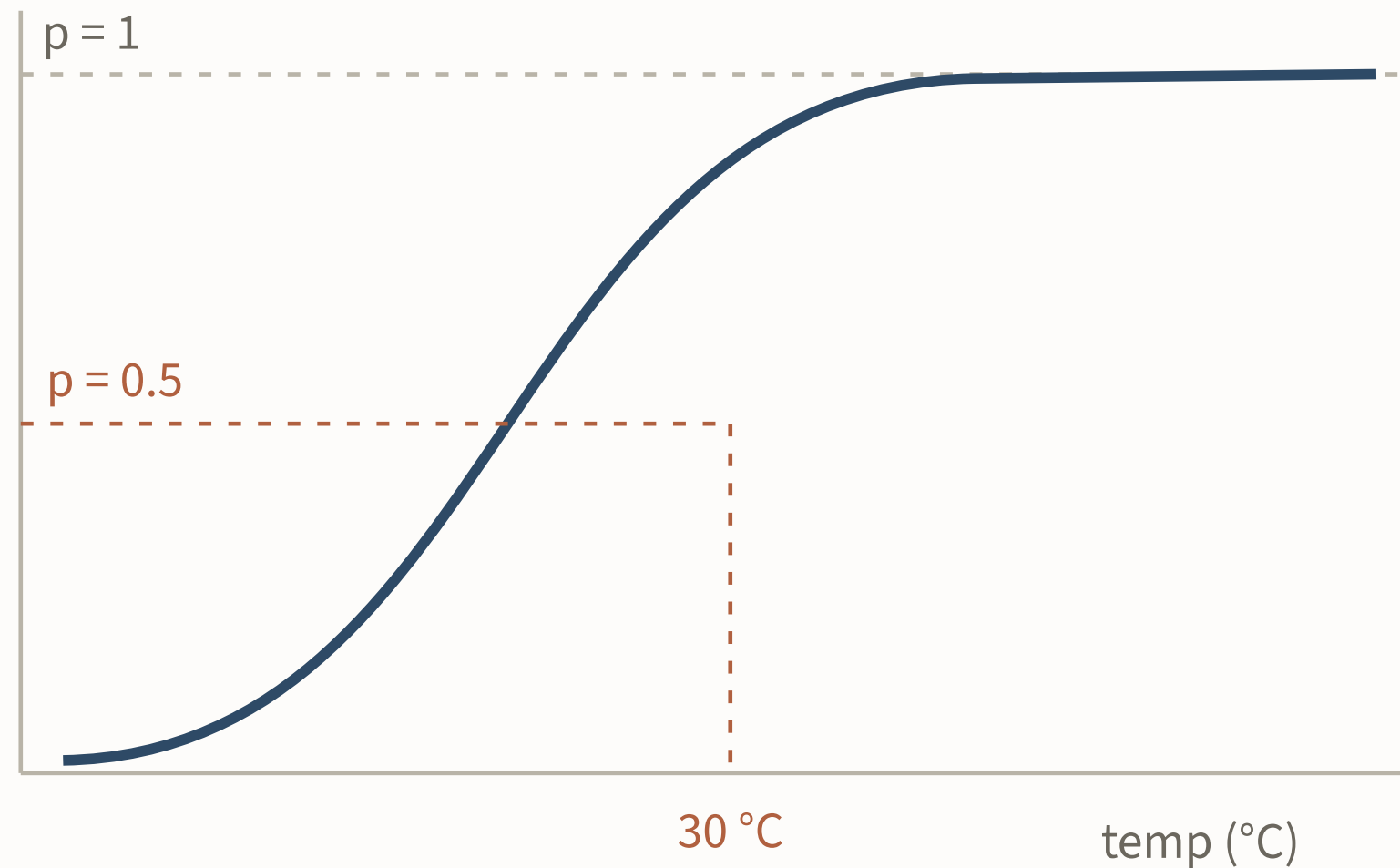
$\log \frac{p}{1 - p}$

lives in $(-\infty, \infty)$ — room for a line at last

The model: put the familiar linear formula on the log-odds scale — $\log\text{-odds}(p) = b_0 + b_1x$. Everything from Part 6 carries over; only the scale changed.

Anchor points: $p = 0.5 \leftrightarrow \text{odds } 1 \leftrightarrow \text{log-odds } 0$. Positive log-odds mean “more likely than not.”

The Logistic Curve



$$p = \frac{e^{(b_0 + b_1 x)}}{1 + e^{(b_0 + b_1 x)}}$$

Solving back for p bends the line into an **S-curve**: flat near 0, steepest at $p = 0.5$, flat again near 1. It can never leave $[0, 1]$.

Reading b_1 : each +1 in x multiplies the *odds* by e^{b_1} . For $b_1 = 0.24$: $e^{0.24} \approx 1.27$ — each degree raises the odds of selling out by ~27%.

Fitting by Maximum Likelihood

There are no residuals to square. Instead we ask: **which b_0 , b_1 make the data we saw most probable?**

Each day contributes its Bernoulli probability — p_i if it sold out, $1 - p_i$ if not. Multiply them all:

$$L(b_0, b_1) = \prod p_i^{y_i} (1 - p_i)^{(1-y_i)}$$

The fitted coefficients **maximize this likelihood** (in practice, its log — sums beat products).

HOW THE COMPUTER SOLVES IT

Unlike least squares, there is **no closed-form formula**. The log-likelihood is a smooth hill; the software climbs it iteratively (Newton's method), stopping when the top is reached — that's the `Iterations` 7 line in the output.

The unifying view: least squares is itself maximum likelihood — for normal errors. One principle, two models: pick the parameters under which your data were least surprising.

```
res = smf.logit("sold_out ~ temp", data=df).fit()  
# Optimization terminated successfully.
```


WORKED EXAMPLE

Predicting a Sell-Out Day

$$\text{log-odds(sold out)} = -7.2 + 0.24 \times \text{temp}$$

Cool day · 20 °C

$$\text{log-odds} = -2.4$$

$p \approx \mathbf{0.08}$ — rarely sells out

Warm day · 30 °C

$$\text{log-odds} = 0$$

$p = \mathbf{0.50}$ — a coin flip

Hot day · 35 °C

$$\text{log-odds} = 1.2$$

$p \approx \mathbf{0.77}$ — stock up

```
smf.logit("sold_out ~ temp", data=df).fit().summary()
```

Note the nonlinearity: the same +5 °C moves p by 0.27 near the middle, but barely at the extremes.

Logistic Regression

- 1 Binary outcome → model the **probability**, and a straight line won't stay in $[0, 1]$.
- 2 The fix: run the familiar line on the **log-odds scale** — logistic regression is linear regression on a stretched ruler.
- 3 Interpret e^{b_1} as an **odds multiplier**, and remember effects on p are largest near $p = 0.5$.

CLOSING

The Road from Data to Inference



Every step reused one idea: an estimate, its standard error, and the question “how surprising is this?”